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Title of Paper (14 pt, Title Case, Boldface)

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Birrul Walidain[1], Nasrullah Idris[1*], Khairun Saddami[2] (12 pt)

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Abstract (12 pt, bold)

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Abstract should be written in both English and bahasa Indonesia. The abstract is written with Times New Roman font size 10, and single spacing. The abstract should summarize the content of the paper, including the aim of the research, research method, and the results, and the conclusions of the paper. It should not contain any references or displayed equations. The abstract should be no more than 200 words.

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Keywords: *up to 5 keywords in English (10 point font, italics); write alphabetically in 5 words.*

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1. **Introduction** (12 pt, bold)

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Asumsi plasma homogen yang selama ini digunakan dalam metode *Calibration-Free Laser-Induced Breakdown* Spectroscopy (CF-LIBS) terbukti tidak memadai untuk mendeskripsikan dinamika plasma rill yang bergradien tinggi[1]. Keandalan metode kuantitatif CF-LIBS mutlak bergantung

pada pemenuhan syarat *Local Thermal Equilibrium* (LTE) sebagai prasyarat utama[2,3]. Dalam kondisi ideal, kerangka termodinamika ini memastikan parameter fundamental plasma, yakni temperatur elektron (T_e) dan densitas elektron (N_e), dapat diekstraksi secara presisi guna mengkuantifikasi komposisi unsur secara langsung, sebuah mekanisme fisis universal yang juga menjadi landasan utama dalam pemodelan campuran ionik pada plasma astrofisika[4]. Sebaliknya, penyimpangan dari kondisi LTE ini tidak hanya mendistorsi akurasi analitik, melainkan mereduksi mekanisme kuantifikasi menjadi tidak valid [2,5]. Mengingat kompleksitas pada karakterisasi plasma yang tervalidasi secara ketat menjadi sangat esensial, khususnya dalam analisis material matriks kompleks yang melekat rentan terhadap anomali ketidakhomogenan plasma pada pengamatan, seperti yang ditemui pada instrumen eksplorasi antariksa [6,7], analisis batuan analog Mars dalam kondisi atmosfer ekstrem[8], karakterisasi in-situ transisi fase logam cair[9], evaluasi abrasi pelindung panas wahana saat memasuki atmosfer[10], maupun pada pemantauan in-situ dinding reaktor fusi nuklir [11].

Meskipun kriteria McWhirter (1965) secara tradisional digunakan sebagai landasan validasi LTE, perumusan klasik ini terbukti menjadi syarat yang diperlukan namun tidak memadai untuk mendeskripsikan dinamika plasma riil[12]. Distorsi paling destruktif pada plasma Adalah dalam bentuk penyerapan mandiri (*self-absorption*), di mana struktur spasial yang berlapis merusak linieritas profil spektral emisi[13]. Ketidakpastian ini semakin diperumit oleh anomali pada profil populasi tingkat energi internal molekuler di mana konfigurasi intensitas transisi vibrasi-rotasi menunjukkan deviasi signifikan dari prediksi LTE tradisional dan menuntut pendekatan diagnostik dua-temperatur pada kondisi non-LTE[14]. Untuk memitigasi anomali spasial tersebut, studi-studi komputasi terkini beralih pada arsitektur Artificial Neural Network (ANN) yang mengintegrasikan model dua zona (two-zone model) [1,15,16]. Namun, sementara pendekatan komputasi data-driven ini berhasil mereduksi interferensi spasial, model-model tersebut masih tertahan pada penyederhanaan kondisi stasioner. Padahal, kajian kinetika teoretis terbaru membuktikan bahwa plasma secara fundamental bersifat sangat transien; laju pendinginan yang ekstrem dan tingginya opasitas radiatif (τ) memaksa ambang batas densitas elektron minimal melompat jauh melampaui prediksi stasioner konvensional[12]. Kondisi ini menyisakan sebuah celah metodologis, di mana arsitektur analitik yang hanya mengandalkan pemecahan ruang spasial tanpa melibatkan filtrasi kinetik berbasis waktu berisiko kuat memvalidasi spektrum non-LTE secara keliru.

Menjawab limitasi komputasi stasioner tersebut, penelitian ini mengusulkan sebuah strategi filtrasi termodinamika dinamis untuk mengevaluasi secara komparatif batas validitas kriteria LTE termutakhir[12] berhadapan dengan asumsi klasik pada sampel matriks kompleks. Berbeda dengan skema konvensional maupun arsitektur probabilitas kecerdasan buatan, studi ini menerapkan pendekatan resolusi waktu (time-resolved) yang ekstensif untuk memetakan dan mengisolasi jendela waktu spesifik di mana kondisi LTE murni benar-benar tercapai[5] selaras dengan optimasi pemrosesan sinyal berbasis waktu pada kuantifikasi material heterogen secara bebas kalibrasi[17]. Sebagai inti komputasi, algoritma analitik penyelesaian Persamaan Transfer Radiatif (RTE) diaplikasikan secara deterministik. Penggabungan pemodelan kinetika dengan penyelesaian RTE ini merupakan standar komputasi mutakhir dalam menangani dinamika plasma optically thick, dan terbukti mutlak diperlukan untuk merekonstruksi spektrum serta meniadakan distorsi penyerapan mandiri pada kondisi non-LTE di mana parameter opasitas radiatif mendikte secara absolut tingkat

absorpsi plasma[3,16,18,19]. Lebih jauh, penyelesaian analitik ini bertindak sebagai generator spektrum sintetis yang absolut krusial untuk melatih arsitektur machine learning secara presisi tanpa kalibrasi empiris[20]. Melalui integrasi pemodelan radiatif ini, penelitian ini berupaya mereduksi ketidakpastian kuantitatif secara fundamental dan menyajikan landasan fisis yang logis dan konsisten, membebaskan komputasi spektral dari batasan stasioner.

The first letter of headings is capitalized and headings are numbered in Arabic numerals. The organization of the manuscript includes **Introduction, Methods** or **Experiment, Results and Discussion, Conclusions** and **References**. **Acknowledgement** (if any) is written after **Conclusion** and before **References** and not numbered. The use of subheadings is discouraged. Between paragraphs, the distance is one space.

Abbreviations/Terms/Notations/Symbols. The use of abbreviations is permitted, but the abbreviation must be written in full and complete when it is mentioned for the first time and it should be written between parentheses. Terms/foreign words or regional words should be written in italics. Notations should be brief and clear and written according to the standardized writing style. Symbols/signs should be clear and distinguishable, such as the use of number 1 and letter l (also number 0 and letter O).

2. Tables (12 pt, bold)

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Tables are written with Times New Roman font size 10 and single spaced below the title of the table. The title of the table is written with font size 9 (boldface) above the table as prescribed in the format given below as example. The table is numbered in Arabic numerals. There is one single space line between the table and the paragraph. The table is placed immediately after it is referred to in the text.

The frame of the table uses 1 font- size line. If the title in each table column is long and complex, the columns are numbered and the notes are given below the table.

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Table 6. Elements in *S. trifasciata* Biosorbent (9 pt, boldface)
(*one blank single space line, 6 point font*)

Element	Before Pb(II) treatment		After Pb(II) treatment	
	Weight (%)	Abundance (%)	Weight (%)	Abundance (%)
C	48.79	56.47	38.90	50.33
O	49.05	42.62	49.39	47.97
Ca	1.23	0.43	2.65	1.03
Al	0.93	0.48	-	-
Pb	-	-	9.06	0.68

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3. Graphic Contents (12 pt, bold)

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Graphic contents are placed symmetrically on the page and there is one blank single space line between the graphic content and the paragraphs. A graphic content is placed immediately after it is referred to in the body of the text and is numbered in Arabic numerals. Caption for the graphic content is written below it and there is one blank single space line between it and the graphic content. The caption is written in font size 9, boldface, and placed as in the example. Between the graphic content with the body of the text there are two blank single space lines.

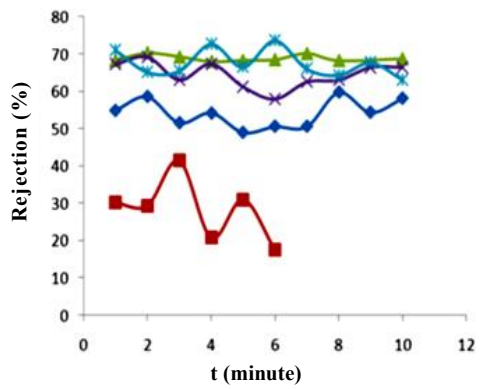

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Figure 1. Value of Chitosan Membrane Rejection, A1, A2, B1, and B2 on the Artificial Wastewater; Chitosan (■), A1 (▲), A2 (X), B1 (✳), B2 (♦) (9 pt, boldface)

For any graphic contents which have been published by another author, the corresponding author must obtain a written permission from that other author and his/her publisher. Include one graphic content which is printed in good quality in a full-size page or scanned graphic content in good resolution in the format as follows {name of file}*.jpeg*, or {name of file}*.tiff*. If the graphic content is in the form of photograph, include one original photograph. The photograph is printed in black and white unless the photograph will appear in color. The author will be charged an additional fee for color printing of more than 1 page. The font face to be used in the graphic content or graph should be the one generally available in each word processor and operational system such as Symbol, Times New Roman and Arial with the font size not less than 9 point.

4. Reaction or Mathematical Equation (12 pt, bold)

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The reaction or mathematical equation should be positioned symmetrically on the column, marked by sequential numbers written on the right corner within brackets. If the writing of equation takes more than one line, numbers should be written on the last line. Letters used as mathematical symbols in the text should be written in italics such as *x*. Equations in the text should be referred to as abbreviations, for example Eq.(1) or Eq.(2).

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$$\mu(n,t) = \frac{\sum_{i=1}^{\infty} 1\,(d_i < t, N(d_i) = n)}{\int_{\sigma=0}^{t} 1(N(\sigma) = n)\,d\sigma} \qquad (1)$$

(one single space, 10 pt)

Equation (1) is obtained using a Style format as the following: Variable: *Times New Romans Italic* and *LC Greek*: *Symbol Italic*. Format size: *Full 10 pt*, *Subscript/Superscript* 8 *pt*, *Subsubscript/Sub-superscript* 6 *pt*, *Symbol* 11 *pt* and *Sub-symbol* 9 *pt*. Elaboration of mathematical equation or formula

need not be written in detail; write only the most important part, the method used, and the final result.

Citations in the text should be written using Arabic numbers and put in order in accordance to what they refer to in the text. Numbers should be written in square brackets such as "... Zhang *et al*. [1] ...." Citations should be written one space away from the words after commas or periods and before colons (:), semicolons (;), and question marks (?). If located at the end of a sentence, citations should be put before periods such as ".... by several researchers [2-3]." All citations should then be written in the right order in the list of references at the end of the text, with the writing procedure as illustrated in the example.

Appendices

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Appendices are used only if really necessary, located before References and after Acknowledgement (if any). If there is more than one Appendix, they have to be written in alphabetical order.

References

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References should be written following the order they appear in the text, using Arabic numbers in square brackets, as seen in the example. References should consist of initials and writers' names, names of journals or titles of books, volumes, editors (if any), publishers and their cities, years of publication, and pages. All writers' names have to be mentioned. Use the abbreviation "Anon" if writers are anonymous. Names of journals should be written using the commonly-used abbreviations.

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